Kingdom of Saudi Arabia
Ministry of Education
Imam Abdulrahman bin Faisal University
College of Science and Humanities-Jubail
Computer science Department

Software Engineering
Term 1 – 2025/2026

Software Project Management Plans

For

HealthyCooking System

SPMP
Version 1.0

Name of Team:

Project Advisor:

Date of Preparation:
14th of October 2025

This Software Project Management Plan (SPMP) was prepared and provided as a deliverable for [Software Engineering Course, 411, Term 1 – 2025/2026], and it will be used by [students and instructors].

This document is based in part on the IEEE Recommended Practice for SPMP Descriptions.

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1. Project Overview

1.1 Purpose, Scope, and Objectives

1.1.1 Background and Purpose

The **HealthyCooking** project unites the principles of *software engineering*, *artificial intelligence (AI)*, and *health informatics* to promote healthier dietary behaviours through an intelligent, user-centred application. Although countless recipe websites and apps exist, users often struggle to find meals that genuinely suit their preferences, allergies, and nutritional goals. This leads to confusion, wasted time, and repetitive eating patterns that neglect health and variety.

HealthyCooking addresses this challenge by offering an integrated, intelligent system that helps individuals discover, analyze, and plan meals efficiently. The application will support decision-making by combining nutritional data with AI-driven personalization to generate context-aware recommendations. The goal is to make balanced eating achievable, practical, and enjoyable.

The purpose of this **Software Project Management Plan (SPMP)** is to establish the managerial and technical framework for developing HealthyCooking. It defines the scope, objectives, resources, constraints, success metrics, and risks that will guide project documentation and planning. This plan follows the **IEEE 1058** standard to ensure systematic planning and consistent documentation. The project will not be implemented at this time; instead, it will concentrate solely on documentation (analysis, design, planning, and risk management). Further work in Application and Web courses, where the implementation phase can be finished, may also be built upon this material.

Prior research confirms that AI-based dietary recommendation systems improve user satisfaction and adherence when combined with structured datasets and adaptive algorithms [1]. Thus, HealthyCooking merges verified nutritional data, responsible AI design, and user-friendly interfaces to encourage sustainable, health-conscious lifestyles.

1.1.2 Scope

HealthyCooking's scope centers on creating complete documentation for a personalized nutrition and meal-planning platform that supports users from meal discovery to execution. Planning, requirement analysis, risk analysis, and design documentation will all be part of this phase, which will not involve system implementation.

Since HealthyCooking can be expanded and used in application and web courses, the project documentation will also explain how this project relates to other academic initiatives.

The project's major functional modules include:

1. **Recipe Discovery:** Enables users to browse a variety of international and regional recipes organized by meal type, cooking time, and dietary needs.
2. **Personalized Recommendation Engine:** Uses a combination of content-based and behavior-based approaches to suggest meals that align with users' tastes and restrictions.
3. **Smart Ingredient Analyzer:** Allows users to enter available ingredients and receive meal options based on what they already have, offering practical substitutions when items are missing.
4. **Weekly Meal Planner:** Automatically generates balanced menus that meet users' caloric and macronutrient goals, adjustable through personal preferences.
5. **Nutritional Database Integration:** Connects with open nutritional sources (such as USDA FoodData Central) to ensure accurate energy and nutrient calculations.
6. **Community and Recipe Sharing:** Encourages users to share their own recipes and experiences, fostering a sense of community and healthy competition.
7. **User Profile Management:** Safely stores information about allergies, dietary goals, and favorite cuisines while respecting privacy and consent.
8. **Analytics Dashboard:** Presents insightful graphs and statistics to help users visualize their nutritional balance and progress over time.

The first documentation version will focus on these eight modules and exclude complex extensions such as grocery-delivery integration or clinical nutrition services. The product documentation will contain all pertinent design schematics and risk assessments and will be accessible to users who understand Arabic and English. The architecture will remain modular and scalable for future expansions.

1.1.3 Objectives

HealthyCooking has multi-layered objectives that reflect academic, technical, and organizational priorities.

Strategic Objectives:
The project seeks to encourage healthier living by providing a convenient and personalized digital assistant for meal planning. It also aims to showcase a fully documented, IEEE-compliant software engineering project that demonstrates the practical application of AI in real-world health contexts.

System Objectives:
HealthyCooking will deliver a prototype capable of generating individualized weekly meal plans and relevant recipe suggestions. The system must respond quickly to user queries, maintaining an average response time of **two seconds or less**, and achieve a **System Usability Scale (SUS)** score of at least 80 in usability evaluations.

Technical Objectives:
The architecture will emphasize modularity, clear data separation, and maintainability. Machine-learning models will adapt to evolving user patterns to refine accuracy over time [3].

Operational Objectives:
The team will ensure ethical and privacy-respecting data management in line with institutional and international regulations. Work will be arranged and fairly distributed, and consistent documentation will guarantee transparency across development phases. To identify possible hazards and specify mitigation techniques, a section on risk analysis will be incorporated.

As noted by [2], the long-term success of AI-driven applications depends on perceived personalization, transparency, and control. Therefore, HealthyCooking will prioritize explainability and user empowerment so individuals understand why specific meals are recommended and can tailor suggestions to their own goals.

1.1.4 Success Metrics and Validation Methodology

The success of HealthyCooking will be measured using a combination of quantitative metrics and qualitative evaluations.

- **Accuracy:** At least 85% of recipe recommendations must correctly meet user constraints such as allergy filters or time limitations.

- **Performance:** Median response time for generating a recommendation should remain below two seconds.
- **Usability:** The prototype should reach a System Usability Scale score of 80 or higher during pilot testing.
- **Engagement:** Users are expected to interact with the system at least three times per week, with a retention rate of 60% or more during the trial period.
- **Behavioral Impact:** Post-study surveys should indicate increased confidence and organization in meal planning [4].

Validation will occur through **unit testing**, **integration testing**, and **scenario-based user evaluations**. Each iteration will refine design and performance metrics. Feedback will guide continuous improvement and ensure that risks are controlled proactively.

1.1.5 Relationship to Other Projects and Integration

HealthyCooking is self-contained but designed for future interoperability with related systems, such as grocery-list applications, wearable health trackers, or calorie-monitoring devices. By maintaining a **loosely coupled architecture**, future modules or integrations can be added without disrupting the main structure. This foresight aligns with long-term scalability and potential research collaborations in health-oriented AI systems.

1.2 Assumptions, Constraints, and Risks

1.2.1 Assumptions

The following assumptions describe the actual working conditions and expectations during the current project period:

- The course instructor will provide timely feedback and evaluations according to the scheduled milestones.
- Each team member will contribute approximately **10 hours per week** to project tasks to ensure consistent progress.
- Pilot users — primarily classmates and volunteers — will provide honest and useful feedback on meal recommendations and usability tests.

1.2.2 Constraints

HealthyCooking operates within several limiting conditions typical of student projects.

- **Time Constraint:** The 15-week semester restricts the number of features and level of testing possible.
- **Budget Constraint:** No direct financial support is provided; development relies entirely on open-source tools.
- **Quality Constraint:** All documentation and outputs must comply with **IEEE 1058** for SPMP and **IEEE 829** for testing.
- **Scope Constraint:** Only the prototype will be delivered; commercial deployment is not included.
- **Technology Constraint:** The system depends on third-party datasets and APIs that may experience downtime or data changes.

1.2.3 Risk Assessment

Risk management is continuous throughout the project lifecycle. The main risks include:

Technical Risks: The recommender model may perform inconsistently across diverse dietary patterns. To mitigate this, the dataset will be periodically expanded and validated for bias [3]. The "cold-start problem," where new users lack data history, will be reduced by providing onboarding questionnaires.

Schedule Risks: Overlapping academic responsibilities may delay milestones. Preventive actions include early task planning and internal deadlines.

Ethical and Privacy Risks: Users' consent and data confidentiality are essential. The team will employ anonymization and obtain explicit permission before storing sensitive data [4].

Human and Organizational Risks: Differences in workload or absence of members could affect progress. A documentation-first strategy ensures continuity and redundancy in critical roles.

External Risks: Temporary API outages or hardware issues may occur. These will be mitigated by maintaining local copies of essential datasets and using cloud storage for backup.

HealthyCooking's design integrates a "human-in-the-loop" principle, ensuring AI suggestions are always reviewable by the user, thereby improving trust and accountability.

1.3 Project Deliverables

Deliverables are defined according to IEEE 1058 guidelines to ensure completeness, traceability, and quality assurance. Each deliverable includes a specific verification method and a measurable outcome.

The first deliverable (**M1**) is this SPMP itself, which outlines the project's objectives, scope, risks, and managerial processes. The second deliverable (**M2**) will be the **Software Requirements Specification (SRS)**, documenting both functional and non-functional requirements derived from user and stakeholder needs. The third deliverable (**M3**) is the **Software Design Specification (SDS)**, detailing architectural structures, data flows, and interface specifications.

Next, **M4**, the **Software Testing Specification (STS)**, will describe the testing procedures, expected results, and metrics for coverage and validation. **M5** will consist of the **Prototype Application**, which integrates all system components in a functional demonstration evaluated by the course instructor. Finally, **M6** will be the **Final Presentation**, summarizing the entire process and showcasing project outcomes.

Each document and artifact will undergo advisor review, peer inspection, or formal walkthrough to confirm completeness, accuracy, and alignment with project standards.

1.4 Schedule and Budget Summary

1.4.1 Schedule

HealthyCooking follows a **hybrid process model** that combines structured planning with agile iteration. The semester is divided into phases, each culminating in a deliverable review.

- **Weeks 1–3:** Topic selection, background research, and preparation of the SPMP.
- **Weeks 4–6:** Requirement elicitation, analysis, and validation.
- **Weeks 7–8:** System architecture and interface design.
- **Weeks 9–11:** Implementation of core modules and integration testing.
- **Weeks 12-14:** Usability testing, refinement, and documentation updates.
- **Week 15:** Final presentation and submission.

Regular review meetings will ensure timely progress tracking. Adjustments to the schedule will be made when necessary to manage academic workload and unexpected delays.

1.4.2 Budget

Although no direct funding is provided, a realistic effort-based budget is maintained to reflect the project's human and resource contributions. The total estimated effort is **980 person-hours**, based on seven team members contributing approximately ten hours per week across **fourteen weeks** (≈ **12,000 SAR** as an educational labor valuation). No licensed software purchases are anticipated; only **standard academic authoring, diagramming, and planning tools** are required for this report and subsequent course deliverables. Hardware consists of **existing personal laptops** and shared cloud storage already available to students. Minor incidental costs—such as internet usage, printing, and submission materials—are expected to be minimal (about **20 SAR per student**, ≈ **140 SAR** total). This cost-efficient approach demonstrates responsible resource management appropriate for an academic environment while retaining professional planning rigor.

1.5 Evolution of the Plan

This SPMP is a living document that evolves alongside the project. Updates will occur after each major milestone or when new risks and constraints are identified. Every revision will be approved by the instructor and communicated to all team members to ensure alignment.

Version control will be maintained through secure document storage. Key project documents—such as the SPMP, SRS, SDS, and STS—will be periodically reviewed to maintain coherence and consistency. Regular peer evaluations and instructor feedback sessions will ensure the plan remains accurate, transparent, and compliant with IEEE standards.

1.6 References

[1] S. Abhari, R. Safdari, L. Azadbakht, and K. B. Lankarani, "A systematic review of nutrition recommendation systems: With focus on technical aspects," J. Biomed. Phys. Eng., vol. 9, no. 6, pp. 591–602, 2019, doi: 10.31661/jbpe.v0i0.1248.

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[3] L. Yang, C.-K. Hsieh, H. Yang, J. P. Pollak, N. Dell, S. Belongie, C. Cole, and D. Estrin, "Yum-me: A personalized nutrient-based meal recommender system," ACM Trans. Inf. Syst., vol. 36, no. 1, pp. 1–31, 2017, doi: 10.1145/3072614.

[4] A. D. Starke, J. Dierkes, G. Arslan, G. A. B. Kasangu, and C. Trattner, "Supporting healthier food choices through AI-tailored advice: A research agenda," PEC Innovation, vol. 6, art. 100372, 2025, doi: 10.1016/j.pecinn.2025.100372.

1.7 Definitions and Acronyms

Acronym / Term	Definition
AI	Artificial Intelligence: Algorithms that learn and generate intelligent recommendations.
API	Application Programming Interface: Channel that enables communication between software components.
CRUD	Create, Read, Update, Delete: Basic database operations.
DBMS	Database Management System: System for managing and retrieving structured data.
GDPR	General Data Protection Regulation: Privacy principles ensuring consent and user control.
IEEE	Institute of Electrical and Electronics Engineers: Developer of key software engineering standards.
MVC	Model–View–Controller: Design pattern separating logic, interface, and data layers.
PII	Personally Identifiable Information: Any information that can identify an individual.
SDS/SRS/STS	Software Design, Requirements, and Testing Specifications used for documentation.
SPMP	Software Project Management Plan defining scope, schedule, and risk control.
SUS	System Usability Scale: A standardized usability evaluation metric (0–100).
UI/UX	User Interface and User Experience: Design principles ensuring clarity and usability.
WBS	Work Breakdown Structure: Framework dividing project tasks into smaller components.
GUI	Graphical User Interface: Facilitates interaction and enhances user experience.
IAU	Imam Abdulrahman Bin Faisal University
PC	Personal Computer
MS	Microsoft

Table 1: Definitions and acronyms

1.8 Document Structure

This SPMP includes **Revision History**, **Table of Tables**, and **Table of Figures**, followed by:

- **Section 1 – Project Overview:** 1.1 Purpose, Scope, and Objectives; 1.2 Assumptions, Constraints, and Risks; 1.3 Project Deliverables; 1.4 Schedule and Budget Summary; 1.5 Evolution of the Plan; 1.6 References; 1.7 Definitions and Acronyms; 1.8 Document Structure.
- **Section 2 – Project Organization:** 2.1 External Interfaces; 2.2 Internal Structure; 2.3 Roles and Responsibilities.
- **Section 3 – Managerial Process Plans:** 3.1 Start up Plan (3.1.1 Estimates; 3.1.2 Staffing; 3.1.3 Project Staff Training); 3.2 Work Plan (3.2.1 Work Breakdown Structure; 3.2.2 Schedule Allocation; 3.2.3 Resource Allocation; 3.2.4 Budget Allocation); 3.3 Project Tracking Plan (3.3.1 Requirements Management; 3.3.2 Schedule Control; 3.3.4 Quality Control; 3.3.5 Reporting; 3.3.6 Project Metrics); 3.4 Risk Management Plan; 3.5 Project Closeout Plan.
- **Section 4 – Technical Process Plans:** 4.1 Process Model; 4.2 Methods, Tools, and Techniques; 4.3 Infrastructure; 4.4 Product Acceptance.
- **Section 5 – Supporting Process Plans:** 5.1 Documentation.
- **Section 6 – Additional Plans:** other plans required for execution and closure.

This structure aligns with the course template and IEEE SPMP guidance.

2. Project Organization

2.1 External Interfaces

The project team selected as a supervisor and representative of the project team, aimed at creating efficient and organized communication pathways with external entities (clients) associated with the project.

Figure 1 demonstrates the critical function of the supervisor as the primary liaison between the project team and external entities, guaranteeing that requirements are clearly defined; client needs are thoroughly understood, and addressed effectively.

The supervisor is also responsible for monitoring the advancement of all phases of the project, offering timely advice and directions, and addressing clients' inquiries and feedback to ensure smooth communication and a high degree of satisfaction among all parties involved. Furthermore, the supervisor guarantees adherence to all regulations and that the project is carried out in a professional and ethical way. It's important to mention that the Culinary Arts Commission is the parent organization, offering support and making certain that the project results are in line with its goals.

Figure 1: External interface

2.2 Internal Structure

The team is led by the Project Leader, who coordinates the efforts among team members and monitors progress to ensure that tasks are carried out efficiently and in accordance with the defined project timeline. The team includes a Planning Manager, responsible for scheduling and task organization; a Design Manager, who oversees prototype design and ensures alignment with project requirements; a Test Manager, responsible for identifying issues and ensuring the quality of the final product; and a Technical Writer, who prepares technical documents and project reports.

Internal communication between team units is maintained through direct meetings to exchange feedback and discuss progress, as well as continuous collaboration via online platforms such as Zoom, which facilitate teamwork and task updates.

Figure 2 illustrates the Internal structure, and **Figure 3** illustrates the organizational structure of the team, showing lines of authority and communication among members. Each member reports directly to the Project Leader, who in turn communicates with the Supervisor to ensure effective coordination and timely decision-making.

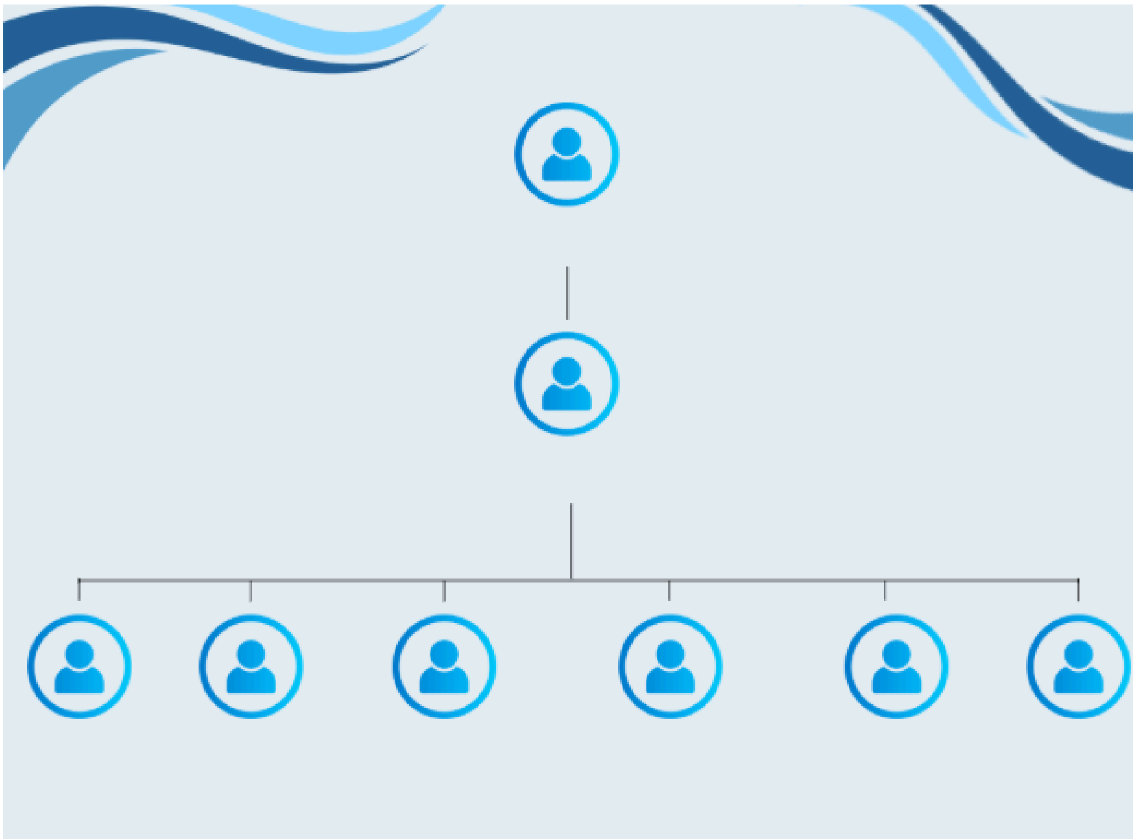

Figure 2: Internal structure

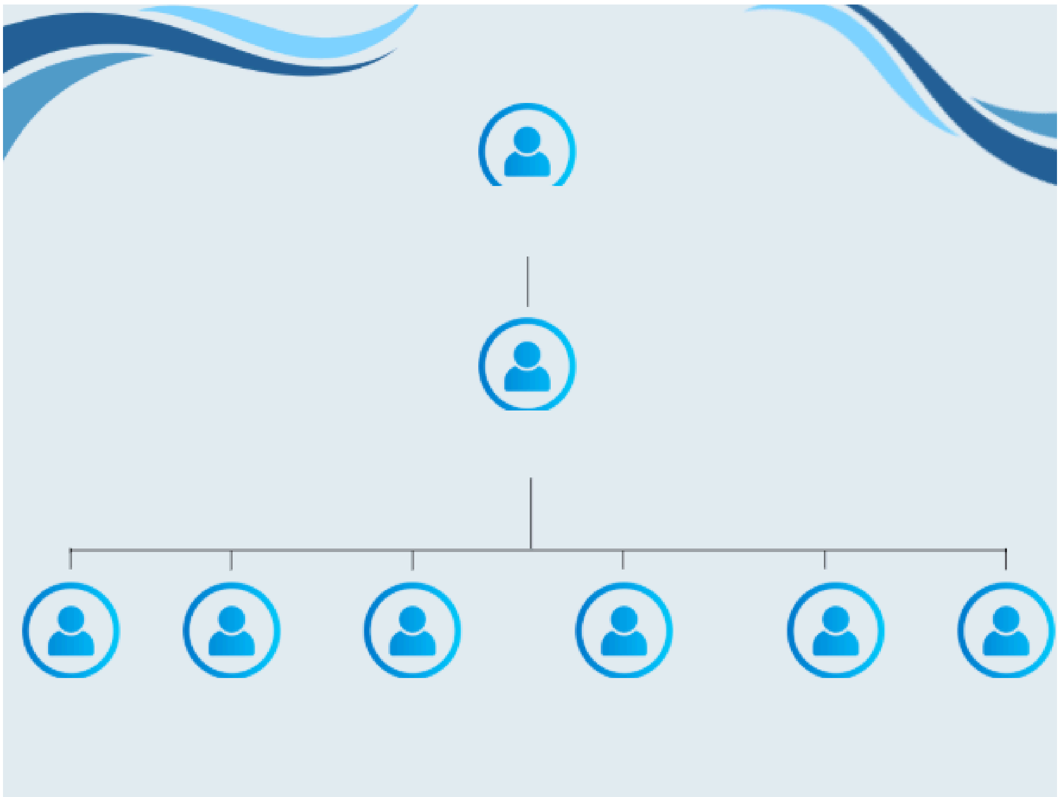

Figure 3: Organizational structure

2.3 Roles and Responsibilities

Establishing roles and responsibilities is a crucial aspect of a project's success, as it aids in structuring the workflow and properly allocating tasks among team members, which in turn ensures that objectives are met both efficiently and effectively. This section is designed to outline the key roles within the project and detail the responsibilities of each member based on their field of expertise. Shown in **Figure 4.**

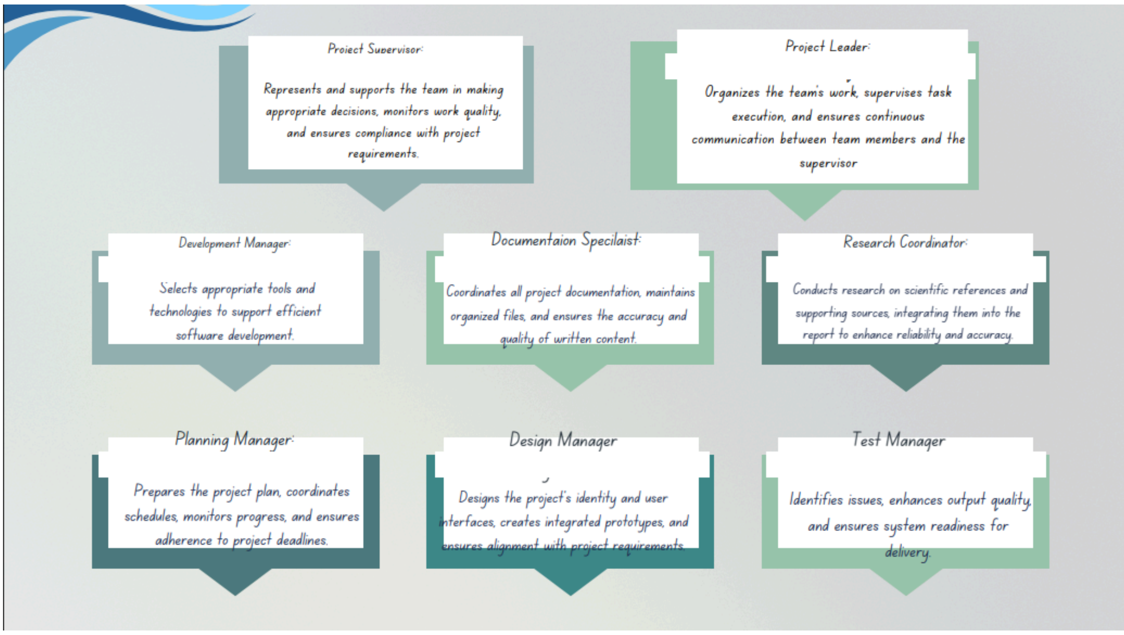

Figure 4: Roles and responsibilities

3. Managerial Process Plans

3.1 Start-up Plan

3.1.1 Estimates

This project is being carried out as a component of the Software Engineering course. It emphasizes applying software management principles in an academic manner rather than developing a real application. There are no real financial expenses, as all the efforts are theoretical and based on our real life experience of preparing, working, and submitting of this project. The project follows a semester timeline outlined in the course project description file that is uploaded in the IAU Blackboard.

Estimated duration per activity:

Defining Project: 4 weeks.

Detailed documentation: 9 weeks.

Prototype Creation: 2 weeks.

Total duration of Activities: 15 weeks.

3.1.2 Staffing

Our team consists of seven students and were chosen by simply knowing each other:

- Team leader; responsible for distributing tasks among us and reviewing our work.
- Team Member.
- Team Member.
- Team Member.
- Team Member.
- Team Member.
- Team Member.

3.1.3 Project Staff Training

No official training session were necessary. However, every team member is taking the course CS411 – Software Engineering every Monday for 3 hours for aroud 15 weeks. Additionally, team members independently went through online tutorials, educational resources, asked for senior's experiments and instructor’s advice.

3.2 Work Plan

3.2.1 Work Breakdown Structure

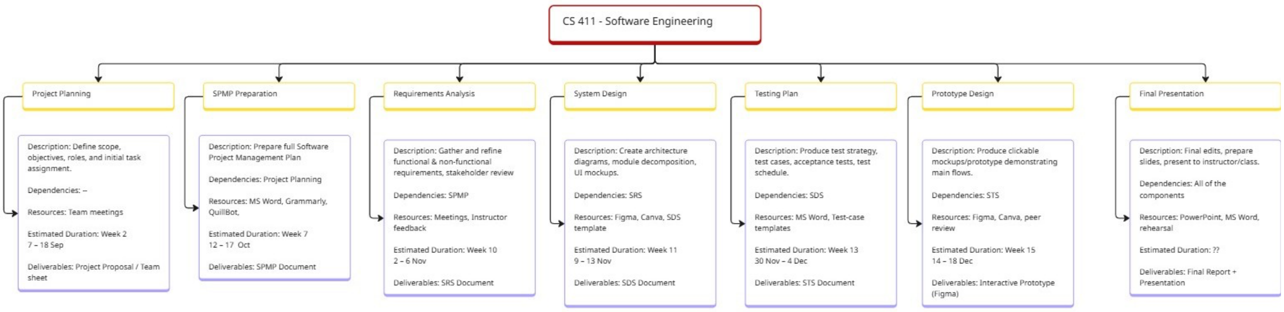

Figure 5: Work breakdown structure

3.2.2 Schedule Allocation

The Project timeline corresponds with the official schedule for CS 411, structured as the Figure 5 shows.

3.2.3 Resource Allocation

All of the project's resources are publicly accessible and personal. Every team member uses a laptop with reliable internet connectivity to contribute. Key software tools include Figma and Canva for design-related tasks, Grammarly and QuillBot for language improvement, and Microsoft Word for writing and formatting documents. Microsoft Teams meetings are used occasionally for talks, Miro for visual planning, and WhatsApp for instant updates when it comes to communication and collaboration. Seven students with different skill levels make up human resources, guaranteeing a range of contributions in testing, design, and writing. Beyond what the team presently has, no extra technical or administrative resources are needed.

3.2.4 Budget Allocation

Since every resource used in this project is either free or privately held, no budget has been allocated. All documentation was finished on personal laptops using free versions of software programs including Microsoft Word, Canva, and Figma. Meetings were held remotely or in college, thus there was no need for outside funding or travel. This strategy guarantees that the project will stay cost-free while preserving a high level of quality and a polished appearance appropriate for an academic context. Total cost: 0 SAR.

3.3 Project Tracking Plan

3.3.1 Requirements Management

To guarantee alignment with the course objectives, requirements are gathered through team discussions and evaluated with the instructor on a regular basis. To ensure consistency, any changes to the requirements are promptly communicated and recorded in shared files that are available to all members. Before approving a modification, the team discusses how it might affect the project's deliverables, schedule, and scope. In addition to supervising these revisions, the team leader ensures that the SRS, SDS, and STS documents are traceable. Documentation and the mapping of requirements to particular design and testing features are supported by tools like Microsoft Word and Miro.

3.3.2 Schedule Control

Through weekly reviews, the team keeps an eye on the project's development and makes sure that every deliverable fits within the syllabus's timetable. Any delays are promptly resolved by redistributing tasks or holding more coordination sessions once progress is compared to planned milestones. To monitor and confirm completion status, tools like Miro WBS, WhatsApp updates, and Microsoft Word checklists are utilized. Prior to submission, the instructor's comments and the project's milestone are used to evaluate the completeness and quality of each document.

3.3.4 Quality Control

Through a consistent organized review and validation procedure, the project maintains its quality. Every deliverable undergoes to be peer review, in which a minimum of one team member assesses the work to guarantee its completeness and accuracy. Tools like Grammarly and QuillBot are used for verification in order to evaluate formatting consistency, grammar, and clarity. The instructor then conducts a final validation to ensure that all entries satisfy the course and academic standards. The team ensures professional quality and uniformity across all project deliverables by following the CS 411 evaluation methodology and IEEE template standards throughout the process.

3.3.5 Reporting

The team uses a combination of tools and reporting procedures to ensure clear and consistent communications. WhatsApp is used to coordinate errors and quick updates, while MS Word is used to prepare and share written reports. Sometimes, when necessary, virtual meetings are conducted using MS Teams to guarantee alignment and development, the team holds daily check-ins and extra meetings prior to each submission deadline. To keep everyone informed a summery update is shared to all members following each meeting. In order to maintain accountability and transparency throughout the project, progress is also verbally shared during class sessions or office hours, and the instructor is consulted whenever needed.

3.3.6 Project Metrics

Project progress is measured using simple academic metrics such as:

Metrics	Frequency	Reporting Process
Document completion percentage	Daily	Reviewed by team leader/team members
On-Time submission rate	Weekly	Compare with syllabus deadlines
Team participation rate	Weekly	Compared by previous meetings
Quality review results	Per-deliverable	Reviewed by team members – Grammarly
Instructor feedback	Per-submission	Recorded in BB or team notes

Table 2: Project metrics

3.4 Risk Management Plan

The *HealthyCooking* project follows a simple process to handle possible risks that might affect the progress or quality of the work.

The team identifies potential problems early, discusses them during meetings, and tries to find solutions before they cause delays.

The three tables below show the main risks that may appear during the project, with their probability, effect, and the way to reduce their impact.

Risk Identification	Affects	Type
R1: Time constraints due to size underestimate	Project and product	Estimation
R2: Staff member absence at critical time	Project	People
R3: Data loss due to internet issues	Project and product	Technology
R4: Devices compatibility issues	Product	Technology
R5: Poor of Academic language	Product	People
R6: Prototype design flaws	Product	Technology
R7: Team disagreements	Project and Product	Organizational

Table 3: Risk identification

Identified Risk	Probability	Consequences
R1	High	Serious
R2	Moderate	Serious
R3	Moderate	Serious
R4	High	Tolerable
R5	Moderate	Tolerable
R6	Moderate	Tolerable
R7	Moderate	Tolerable

Table 4: Risk analysis

Identified Risk	Strategy
R1	Follow a clear schedule, divide tasks early, and avoid last-minute work.
R2	Team should do more overlap work, and should understand each other's job.
R3	Save work regularly, enable auto-save, and try to store files on OneDrive.
R4	Use one software to final review the documents before submitting the files.
R5	Professional members check the report with writing tools before submitting.
R6	Maintain version control to track and correct design changes efficiently.
R7	Solve disagreements by group discussion or voting to reach a fair decision.

Table 5: Risk planning

The team will keep checking these risks throughout the project and update the plan if new problems appear.

3.5 Project Closeout Plan

At the end of the *HealthyCooking* project, the team will follow a simple process to close the project in an organized way.

After completing all tasks, the members will review the system and report to ensure that everything meets the project objectives.

The main steps of the closeout plan are:

1. **Final Review:**

   Each member will check their assigned part and make sure all project goals are achieved successfully.

2. **Final Report and presentation:**

   The group will prepare the final report and presentation that summarizes the project results and challenges faced during development.

3. **Project Closure Confirmation:**

Once all required documents and the presentation are submitted and approved, the project will be officially marked as completed.

This plan ensures that the project ends smoothly and that all work is properly reviewed before final submission.

4. Technical Process Plans

4.1 Process Model

In HealthyCooking project, a number of documents will be written and submitted as an incremental delivery, with each major report delivered and reviewed before proceeding to the next. The incremental process model is illustrated in the following figure.

Information and work products will flow between team members and activities, with each report shared and reviewed before moving to the next stage.

After the submission and approval of the project suggestion and proposal idea in the third and fourth weeks of this current semester.

First of all, with the help of this document, the Software Project Management Plan SPMP, all start-up, work, project tracking, risk management, project closeout, technical process, supporting, and additional plans would be defined and submitted by the end of week 7.

Next, a status report will be prepared in the eighth week to track the progress and ensure alignment with the project schedule.

Then, the Software Requirement Specification SRS document will be considered as the second main document to be delivered in this project, specifying the general and detailed ideas, targeted users, features, and constraints of our HealthyCooking system. It will be produced by the project team in the 10th week of this course.

After that, a Software Design Specification SDS document will be developed at week 11, describing the system requirements, architecture, modules, and database and interface designs.

Next, a second status report will be presented in week 12, to inform the instructors and advisors about the progress of the team work.

For the last document, which is the Software Test Specification STS, test cases and environment will be represented, besides the acceptance criteria. And that would be developed by the end of week 13.

Each major report will go under internal review before submitting it to the instructor for approval. And each submitted document will serve as a baseline for subsequent project activities.

Lastly, the team will deliver the prototype of the HealthyCooking system and will present a presentation of the project in week 15 of this semester as the closure activity of all project deliverables.

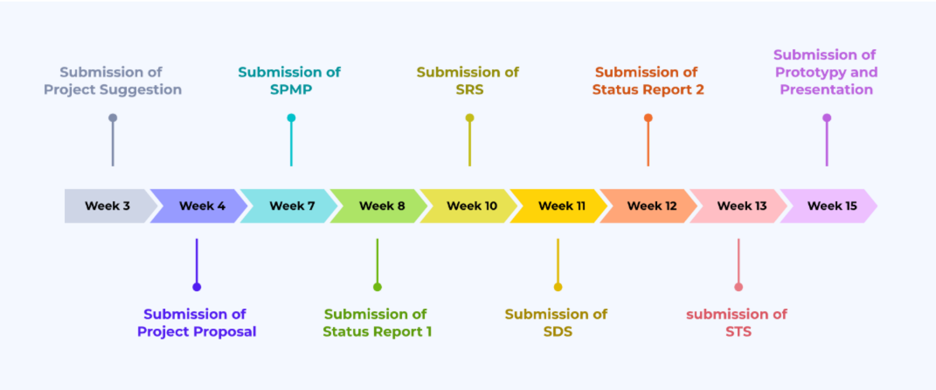

Figure 6: Project incremental process model for reports delivery

4.2 Methods, Tools, and Techniques

The project follows the incremental delivery process model, since the team will deliver a document every week or two, and will wait for approval from the instructors and advisor. And if there are any changes needed in the document, the team will prepare another version of that document.

Team members will write a checklist for each major document to divide the document parts among them.

The Google Calendar application will be used to create student tasks and to specify the deadline date of each document.

For writing documents, Microsoft Word application would be used. And for the figures, the Canva website will be utilized as a tool for creating them.

All deliverables follow standard software engineering documentation practices. Project documents will follow IEEE standards and institutional templates.

Each student member will work individually on their specified part, then the leader of the group will review the document to see if there are any errors before submitting the file. If there are any, errors will be modified by the same member who wrote that part.

Every document version will be labled clearly and will be stored in shared OneDrive shared file.

Furthermore, the team will use Grammarly and QuillBot as tools to review documents as a last step to make sure that the documents are well written.

The team will use the Figma application to make the prototype of this HealthyCooking system project, as a last step of delivery.

For delivery, documents would be submitted electronically through the IAU university's Blackboard website.

4.3 Infrastructure

For the hardware, team members' own PCs and laptops would be utilized, each with a compatible operating system, mostly Windows and macOS. Local networks will be used for searching, communication, documenting in shared file, and submitting the work. And as mentioned above, several software applications will be considered using, such as Microsoft word, Grammarly, and Figma. All work will be conducted remotely using shared platforms as OneDrive and Blackboard. No dedicated physical facilities are required. And the team will follow the university's policies for data security and file management.

4.4 Product Acceptance

For advisors' acceptance, they would be reviewing the documents after the team leader submission. Then they would give feedback and grade the work of each delivery based on the posed rubric in the Blackboard. The rubric used in Blackboard, agreed by instructors and the team members will serve as the formal acceptance criteria.

Objective criteria for determining acceptable work are about clarity, organization, IEEE compliance, content completeness, references, and supporting materials of the documents.

If the work is not expected by the advisors, another version will be created, and the final version of these documents should be approved.

5. Supporting Process Plans

5.1 Documentation

This section explains the supporting process plans that will guide the Healthy Cooking Project to ensure integrity, good quality, and coordination throughout all phases of development.
These plans outline how our team will manage documentation consistency, maintain good quality, track and verify deliverables, review and resolve issues, and continually improve our team's performance.

The project documentation will adhere to the IEEE 1058 standard and the university's official SPMP template. We will store and save our files on OneDrive so that the entire team can access the documents. Our files will be uploaded to Microsoft Word.

The validation process includes reviewing all documents such as the SRS system, design document, and test plan, to ensure their accuracy, completeness, and alignment with the project objectives.

The validation process ensures that the proposed system meets user needs by assessing whether the proposed and described features, such as finding available recipes, filtering our ingredients, AI-based ideas and suggestions, and tracking nutrition and healthy eating achieve the desired benefit of helping users easily plan healthy meals.
Our team also conducted reviews and meetings to identify errors or inconsistencies prior to work.In order to keep track of our work, progress, and updates including objectives, information, ideas, and quality assurance and make sure that everything was in line with our concept criteria, our team also met once a week.

Review before Submitting	Prepared By	Template	Deadline	Document
The Lader	All Team Members	University Template	Week 4	Project Proposal
The Lader	All Team Members	IEEE 1058	Week 7	SPMP (Software Project Management Plan)
The Lader	All Team Members	IEEE 830	Week 8	Status Report 1
The Lader	All Team Members	IEEE 1016	Week 10	SRS (Software Requirements Specification)
The Lader	All Team Members	University Template	Week 11	SDS (Software Design Specification)
The Lader	All Team Members	IEEE 829	Week 12	Status Report 2
The Lader	All Team Members	Internal Template	Week 13	STS (Software Test Plan)
The Lader	All Team Members	Course Template	Week 15	Final Presentation & Prototype

Table 6: Details about project documents

We created our schedule, which also displays and includes all the project work and documentation that will be completed and created for our healthy cooking project. This schedule clearly outlines everything we have

Here we have these standards that must be adhered to, clear and specific standards (IEEE and university) to include quality and clarity.

In our team, we will save all versions in a Google Drive folder that will be shared between the team members, and the leader will check all reviews and versions before submitting.

6. Additional Plans

6.1 System Operation and User Support

Several additional plans have been developed as part of the project documentation to ensure that the HealthyCooking project remains well-organized and clearly structured. These plans outline how different aspects of the project should be managed and reviewed. They include a plan for maintaining and updating the project materials, a plan for reviewing and improving written documents, and a plan for documenting privacy and security guidelines to ensure responsible handling of user-related information in future development stages.

In addition, the project includes a plan for collecting feedback and documenting suggested improvements that may guide future updates. Each of these written plans serves as a reference for future contributors, helping to maintain consistency, quality, and clarity across all project stages. Together, these documented plans ensure that HealthyCooking can continue to grow and adapt while maintaining its organized and professional structure.